
UNIT 2 RELATIONS

Structure

- 2.0 Introduction
- 2.1 Objectives
- 2.2 Relation
 - 2.2.1 Definition, Notation and Illustrations
- 2.3 Properties of Relations
- 2.4 Equivalence relations, and Partition of a Set
- 2.5 Partial Order Relation, Partially Ordered Set
 - 2.5.1 Applications of Equivalence Relation and Partition of a Set
 - 2.5.2 Partial Order Relation, Partially Ordered Sets
- 2.6 Operations on Relations
 - 2.6.1 Set-type Operations on Relations
 - 2.6.2 Mapping-type operations: inverse of operation, composition
 - 2.6.3 Closure Operations
- 2.7 Summery
- 2.8 Answer to Check Your Progress
- 2.9 References and Further Readings

2.0 INTRODUCTION

In Unit 1, we have already talked of set-relations (i. e., relations on sets)—like (subset), and (proper subset). Also, we encounter several situations in day-to-day life to judge number-relations—like $<$ (less than), $>$ (greater than), and $=$ (equal to)—between pair of numbers. In everyday life, we talk of relations like of 'father to daughter', 'mother to son', 'sister to brother', 'teacher to student', and 'employer to employee' to name a few.

In most of such cases, there is a sense of direction/order from the first to the second in the sense that if X is mother of Y , then Y is not mother of X , i.e., X and Y cannot be exchanged in the relation of is-mother-of. Similarly, for the relation of is-less-than ($<$), we have $4 < 5$, and we cannot exchange roles of 4 and 5 for the relation to be true. Except in special cases of relations like equal-to ($=$) in numbers, or the relation of sibling (persons, male or female, having same pair of parents) among human beings, the order of arguments involved in a relation, in general, has to be taken into consideration.

Also, a relation need not be between two similar sets, as discussed above: between two sets of sets, between two sets of numbers, or between two sets of human beings. It may be between two different types of sets. For example, the relation of age-of-person is from a set of human beings to a set of numbers. This leads us to the concepts 'ordered set' and 'cross-product of two sets', which are discussed next.

a) Ordered Set: Earlier, in Unit 1, it was pointed out that in a set, order of occurrence of elements is irrelevant:

For example, $\{1, 2, 3, 4\} = \{4, 2, 1, 3\} = \{3, 2, 4, 1\} = \dots$

However, there are situations, as mentioned just above, in which the change of order of occurrence of the elements, changes the situation. For example, in coordinate

geometry (Fig. 3.1), the location of the point (2, 5) is different from the location of the point (5,2).

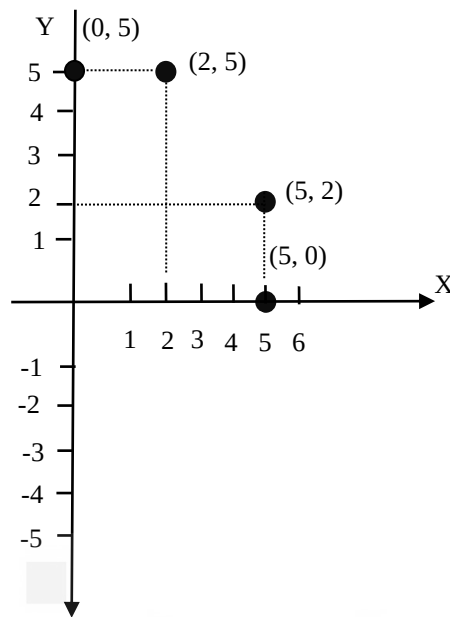


Fig. 2.1: Location of Points in Co-ordinate Geometry

Such a set, in which order of occurrence of elements is significant, is called an **ordered set**. In the notation of ordered sets, the elements are enclosed between parentheses, instead of braces/**curly-brackets**. Thus

$(a, b) \neq (b, a)$, but $\{a, b\} = \{b, a\}$.

b) **Cross-product of two sets X and Y** : for $X \neq \emptyset \neq Y$, denoted by $X \times Y$ is the set of all ordered pairs (x, y) with x belonging to X and y belonging to Y , i. e.

$$X \times Y = \{(x, y): x \in X, y \in Y\}.$$

If $x = \emptyset$ or $y = \emptyset$ then $X \times Y = \emptyset$

In this unit you learn more on relations.

2.1 OBJECTIVES

After going through this unit, you should be able to:

- define relation,
- explain concepts of relation,
- use the notation for a relation,
- explain different properties like reflexivity, symmetry and transitivity of relations,
- apply the concepts of equivalence relation, equivalence class, and partition of a set,
- identify partial order relation, and partially ordered set,
- draw Hasse diagram for a partially ordered set, and
- apply operations, like composition, inverse and union, to relations.

2.2 RELATION

The concept of 'relation' is one of the few fundamental concepts of mathematics. We begin the discussion with its formal/ mathematical definition and illustrations through some well-known examples.

2.2.1 Definition, Notation and Illustrations

In Mathematics, a relation from a set X to a set Y is defined as a **subset** of the cross-product $X \times Y$. Earlier mentioned relations like is-mother-of or is-less-than can be discussed as subsets of cross-products of some sets as follows.

Notation (in Mathematics)

If Sheela is the mother of Shivam, then the fact may be expressed as Sheela **is-mother-of** Shivam.

In mathematics, it is preferred to denote the fact as

(Sheela, Shivam) $\in$ is-mother-of,

where the relation “**is mother of**” is considered as **the set** of all the pairs (m, c) , for which m “is mother” of c .

In one of the methods of mathematical shorthand, **is-mother-of** is denoted in set notation as:

is-mother-of = $\{(m, c): m \in X, c \in Y, \text{ and } m \text{ is mother of } c\} \subseteq X \times Y, \dots(i)$

where $X = Y = \text{set of human beings.}$

(The above form (i) is called the **set-builder form**, in which the name of the property viz. **is-mother-of** is used)

Then the fact Savitri is not the mother of Mohan, is denoted as

(Savitri, Mohan) $\notin$ is-mother-of.

Similarly, the relation 'is-less-than' ($<$) on N , may be considered as a subset of a cross-product by taking $X = Y = N = \{1, 2, 3, \dots\}$ —as follows

is-less-than = $\{(1, 2), (1, 3), (1, 4), \dots, (2, 3), (2, 4), (2, 5), \dots, (3, 4), (3, 5), (3, 6), \dots\}$, or more systematically as is-less-than = $\{(1, 2), (1, 3), (1, 4), (2, 3), (1, 5), (2, 4), (1, 6), (2, 5), (3, 4), (1, 7), (2, 6), (3, 5), (1, 8), (2, 7), (3, 6), (4, 5),$

$(1, 9), (2, 8) \dots\} \subseteq N \times N \dots(ii)$

Thus, $(1, 4) \in \text{is-less-than}$, and $(4, 1) \notin \text{is-less-than}$.

(The above form (ii) is called **roster form**, in which each pair for which property holds is mentioned explicitly instead of the name of the property is-less-than)

Relations

The set-builder form of “is-less-than” is given by $\text{is-less-than} = \{(x,y): x,y \in \mathbb{N}, \text{ and } x < y\}$

Arguments of a relation: the values supplied to a relation are called arguments of the relation. The relations discussed just above involve exactly two arguments: two human beings, two numbers, or two sets. These relations are known as **binary relations**. However, some relations may involve just one argument, e. g., is-male or is-female: for example,

is-male (Mohan) is true, and **is-female (Mohan)** is false.

A **unary relation** is also called a property of its argument; e.g., is-male is a property of Mohan, is-female is not a property of Mohan; similarly, is-prime-number is a property of 19, is-composite-number is not a property of 19.

Some relations may involve more than two arguments; e. g. the relation is- integer-strictly-between involve three integers, such that 4 is-integer-strictly- between 7 and 2 is True. But 4 is an integer strictly between 7 and 9 is False. Also, 4 is an integer strictly between 1 and 3 is also False. We may express the relation in set-builder form as $\text{is-integer-strictly-between} = \{(x, y, z): \text{either } y < x < z, \text{ or } y > x > z\}$. This relation involves three arguments, and such relations are called **ternary relations**.

Similarly, the relation **are-parents-of** involves three arguments, and may be defined as $\text{are-parents-of} = \{(x, y, z): x, y, z \text{ are human beings; and } x \& y \text{ are parents of } z\}$. Similarly, in economics, the (item, cost-production, sales-price) is a ternary relation.

Similarly, we may have relations involving four, five or more arguments. For example, the relation of the timetable of a high school class may involve four arguments as follows.

Time-table = $\{(class, subject, subject-teacher, room-number): class \in \{VI, VII, VIII, IX, X\},$

$subject \in \{Science, Maths., Hindi, English, \dots\},$

$subject-teacher \in \{Ms. Geeta, Mr. John, Mr. Abdul, \dots\},$

$room-number = \{1, 2, 3, \dots\}\}.$

This is an example of **quaternary** relation. In general, a relation R may involve n sets, say $X_1, X_2, \dots X_n$, then $R \subseteq X_1 \times X_2 \times \dots \times X_n$, and R is called **n-ary relation**.

However, most of the time, we will be discussing only relations, like is-mother-of or is-less-than, each of which requires exactly two arguments. A relation which involves two arguments is called a **binary relation**.

Unless we explicitly state otherwise, the term 'relation' will stand for '**binary relation**'.

Definition: A relation between two sets A and B is a subset of $A \times B$. Any subset of $A \times A$ is a **relation on the set A**.

For instance, if $A = \{1,2,3\}$ and $B = \{p,q\}$, then the subset $\{(1,p),(2,q),(2,p)\}$ is a relation on $A \times B$. And $\{(1,1), (2,3)\}$ is a relation on A .

Also, $R = \{(x,y) \in \mathbb{N} \times \mathbb{N} : x > y\}$ is a relation on $\mathbb{N}$, the set of natural numbers, since $R \subseteq \mathbb{N} \times \mathbb{N}$.

If $R \subseteq A \times B$,. We write xRy if and only if $(x,y) \in R$ (xRy is read as ‘ x is related to y ’).

Theorem 2: The total number of distinct relations between a finite set A and a finite set B is 2^{mn} , where m and n are the number of elements in A and B , respectively.

For example, $R_1 = \mathbb{N} \times L$, where L is the set of straight lines; in this relation, we can give different orderings of the straight lines.

If the relation $R_2 = \{1,2,3\} \times \{1_1,1_2\}$, then line 1_1 and 1_2 can get three different ordering.

‘Proof: The number of elements of $A \times B$ is mn . Therefore, the number of elements of the power set of $A \times B$ is 2^{mn} . Thus, $A \times B$ has 2^{mn} different subsets. Now every subset of $A \times B$ is a relation from A to B , by definition. Hence the number of different relations from A to B is 2^{mn} .

Some more definitions

- **Domain, Codomain & Range of a relation:** For a binary relation R from X to Y , i. e. $R \subseteq X \times Y$, the set X is called the domain of R and the set Y is called the codomain of R . For example, consider the relation $R = (\text{Person}, \text{Number})$, number denotes a 10-digit mobile number. Then the domain of R is the set of all persons, and the codomain of the relation R is the set of all 10-digit natural numbers. The set of all 10-digit numbers, which are used as 10-digit mobile numbers of some persons, constitutes the range of R .
- **Image of an element of the domain:** For the relation $R \subseteq X \times Y$, for $x \in X$ (domain), and $y \in Y$ (codomain), if $(x, y) \in R$, then an image of x under relation R is y . For example, If Sheela is mother of Shivam, then under the relation is-mother-of, one of the images of Sheela is Shivam.
- **Relation on a set X :** For a binary relation R , if codomain = domain (i. e., R is from X to X), then R is said to be a relation on X .
- **Onto relation:** For a binary relation R from X to Y , we know $R \subseteq X \times Y$. It is not necessary that every element of Y should occur as second component of some ordered pair in R . Consider the relation of (Person, Number), number stands for 10-digit mobile number. Then, it is not necessary that every 10-digit number is a mobile number of someone. Thus, (Person, Number) is not an onto relation. But the relation “is-mother-of” is onto, because, if P is a person, then P must have a mother, say M , Thus, P occurs as second component of (M, P) .
- **One-to-one relation:** A binary relation R from X to Y is said to be one-to-one if any two distinct elements of the domain cannot have the same image. One of the examples for one-to-one relation is (Person, Mobile- Number); no

Relations

two different persons can have the same mobile number.

- **Empty Relation:** A relation R from a set X to a set Y is called an empty relation if no element of X is related under R to any element of Y , i. e. $R = \emptyset \subset X \times Y$. For example, $X = \text{set of decimal digits} = \{0, 1, 2, 3, \dots, 9\}$, and the relation R is defined on X as $R = \{(a, b): a - b = 10\}$. This is an empty relation, as no pair (a, b) of decimal digits satisfies the condition $a - b = 10$.
- **Universal Relation:** A relation R from a set X to a set Y is called a universal relation if each element of X is related to every element of Y , i.e., $R = X \times Y$.
- The empty relation and the universal relation are sometimes called trivial relations.
- **Notations re-visited:** In general, for R , a relation on a set X , the fact that x is related to y under R may be denoted by either of the notations (i) $(x, y) \in R$, or (ii) $x R y$. Apart from the above-mentioned set notations for relations, two more forms are popular, (i) Graphs and (ii) Matrices. Fig.3.2 explains how to obtain graphic representation of a relation, say,
- $R = \{(1, a), (1, b), (2, a), (3, a), (4, b), (4, c), (4, d)\}$ from the set $X = \{1, 2, 3, 4\}$ to the set $Y = \{a, b, c, d\}$

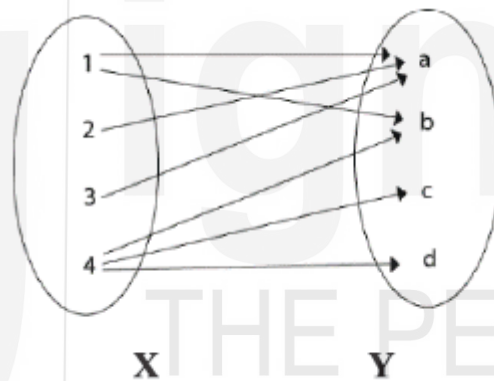


Fig. 2.2: Graphic Representation of a Relation

Fig. 2.3 given below explains how to obtain a matrix representation of a relation, say,
 $R = \{(1, a), (1, b), (1, c), (2, b), (2, c), (2, d)\}$ from the set $X = \{1, 2, 3, 4\}$ to the set $Y = \{a, b, c, d\}$

$$\begin{array}{c}
 \begin{array}{cccc}
 & a & b & c & d \\
 \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \end{array} & \left\{ \begin{array}{cccc}
 1 & 1 & 1 & 0 \\
 0 & 1 & 1 & 1 \\
 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0
 \end{array} \right\}
 \end{array}
 \end{array}$$

Fig. 2.3: Matrix Representation of a Relation

For representing a relation R on a set X , i. e. when domain of R = codomain of R , a directed graph may also be used. For example, the relation $R = \{(1, 1), (1, 2), (2, 2), (3, 1), (3, 3)\}$ on the set $X = \{1, 2, 3\}$, may be represented by the following directed graph (Fig. 3.4).

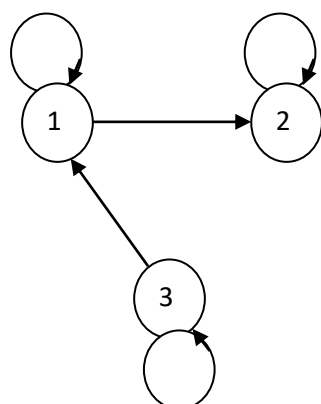


Fig. 2.4: Directed Graph Representing a Relation R on a Set X

In the next subsection, we discuss relations on (set of) relations (rather, we will discuss mainly properties (i.e. binary relations) of relations. In the subsequent subsection, we discuss operations on (set of) relations.

2.3 PROPERTIES OF RELATIONS

We will discuss properties of only for those relations R for which the domain and range is the same set, say X , i.e., the relations on a set, say X . Unless mentioned otherwise, all relations are on the set X (fixed for all the relations under consideration). The discussion revolves around three important properties of relations on a set, viz. (i) reflexive, (ii) symmetric, and (iii) transitive. Other relations to be discussed are directly or indirectly modifications of these properties.

1) Reflexive, Not-Reflexive, Anti-Reflexive/Irreflexive

a) A relation R on set X is said to be reflexive, if for every $a \in X$, then $(a, a) \in R$. For example, each of the following is a reflexive relation

i) the relation of equality ($=$) on $N = \{1, 2, 3, \dots\}$

ii) The universal relation on X , is reflexive.

b) by definition under (a) above, R is not reflexive on X , if for at least one $a \in X$, we have $(a, a) \notin R$. For example,

i) Let $X = N = \{1, 2, 3, \dots\}$. Define $(a, b) \in R$, if and only if the product $(a \times b)$ is an even number. Then $1 \times 1 = 1$ is not even, therefore $(1, 1) \notin R$. Therefore, R is not reflexive on N . Note that there are other elements like 3, for which $(3, 3) \notin R$, but we need not discuss other cases, as just one example of $(1, 1) \notin R$ is sufficient to establish R as not reflexive.

Relations

ii) The empty relation on a set $X \neq \emptyset$ is not reflexive, because for $a \in X$, then $(a, a) \notin \emptyset$ (because $\emptyset$ has no element).

iv) We may construct our own simple counter-examples, e. g. Let $X = \{1, 2, 3\}$, and $R = \{(1, 1), (1, 2), (1, 3), (3, 3)\}$. Then as $(2, 2) \notin R$, R is not reflexive.

c) **Anti-reflexive/irreflexive:** A relation R on set X is said to be anti-reflexive (or irreflexive), if for every $a \in X$, we have $(a, a) \notin R$. For example, the relation

i) is-less-than ($<$) on numbers is anti-reflexive, as $x < x$ is false for every number x , i. e. $(x, x) \notin$ is-less-than

ii) is-mother-of on the set of human beings is anti-reflexive, for every human x , $(x, x) \notin$ is-mother-of

iii) $R = \{(2, 3), (3, 1), (2, 1)\}$ on the set $\{1, 2, 3\}$ is anti-reflexive.

Remarks 1:

a) Difference between not-reflexive and anti-reflexive

For a relation R to be not-reflexive, we require for some $a \in X$, (a, a) fails to be in R .

For a relation R to be anti-reflexive, we require for every $a \in X$, (a, a) fails to be in R .

b) Every anti-reflexive relation is a not-reflexive relation, but a not-reflexive relation may not be anti-reflexive.

If R is anti-reflexive on set X , then, by definition, for every $a \in X$, $(a, a) \notin R$ implies for at least one a , $(a, a) \notin R$ implies R is not reflexive.

Some examples:

i) $R = \{(2, 2), (3, 3), (2, 1)\}$ on the set $\{1, 2, 3\}$ is a not-reflexive relation, as $(1, 1) \notin R$. But it is not anti-reflexive, as $(2, 2) \in R$.

ii) The relation is-brother-of on the set of all human beings is not reflexive, because, if x is a female human being, then $(x, x) \notin$ is-brother-of. But the relation is not anti-reflexive; also, as for each male x , we have $(x, x) \in$ is-brother-of, contradicting the requirement of anti-reflexivity.

c) Just by changing the underlying set of a relation, its property may change.

For example,

i) For the relation “is-brother-of”, if

a) Underlying set $X =$ set of all males, then the relation is reflexive

b) Underlying set $X =$ set of all females, then the relation is anti-reflexive

ii) For the relation R on integers, such that $(x, y) \in R$ if and only the product $x \times y$ is even, then if

- a) Underlying set X = set of **all even integers**, then the relation is reflexive
- b) Underlying set X = set of **all integers**, then the relation is not reflexive, and also not anti-reflexive
- c) Underlying set X = set of **all odd integers**, then the relation is anti-reflexive

2) Symmetric, Not-Symmetric, Anti-Symmetric, Asymmetric

- a) A relation R on a set X is **symmetric** for $a, b \in X$, if $(a, b) \in R$, then $(b, a) \in R$.

Note: $\emptyset$ relation is a symmetric relation.

For example, each of the following is a symmetric relation:

- i) the relation of equality ($=$) on $N = \{1, 2, 3, \dots\}$.
- ii) The relation of is-congruent-to on the set X of all triangles
- iii) the relation of is-sibling-of on the set of all human beings.

But the relation less-than-or-equal-to ($\leq$) is NOT symmetric, though $\leq$ is reflexive on numbers.

b) By definition (a) above, a relation R on a set X is **not symmetric** if there is some pair $a, b \in X$, with $(a, b) \in R$, but $(b, a) \notin R$. For showing not-symmetric, we need to show just one instance violating the condition of symmetry.

Some examples are:

- i) The relation R of less-than-or-equal-to ($\leq$) on integers is NOT symmetric, as $(3, 4) \in R$, but $(4, 3) \notin R$
- ii) Relation R of is-mother-of on the set of all human beings is not symmetric
- iii) Relation $R = \{(1, 1), (2, 2), (3, 3), (1, 2)\}$ on the set $\{1, 2, 3\}$ is NOT symmetric, as $(1, 2) \in R$, but $(2, 1) \notin R$. We may note that R is a reflexive relation

c) **Antisymmetric:** a relation R on a set X is anti-symmetric, if we have $(a, b) \in R$, and $(b, a) \in R$ —for two elements $a, b \in X$ —then we must have $a = b$.

Examples:

- i) The relation of “is less than or equal to” on integers is antisymmetric
- ii) On the set $X = \{1, 2, 3\}$:
 - a) Each of the relations $\{(1, 1)\}$, and $\{(1, 1), (2, 2)\}$ is both symmetric and antisymmetric.
 - b) The relation $\{(1, 2), (1, 3)\}$ is antisymmetric, but not symmetric.

Remarks 2:

a) Difference between not-symmetric and anti-symmetric: For a relation R to be not-symmetric, we require for some pair $a, b \in X$, if $(a, b) \in R$, then (b, a) fails to be in R . For a relation R to be anti-symmetric, we require for every pair $a, b \in X$, if $(a, b) \in R$, and $a \neq b$ then (b, a) fails to be in R .

b) Every anti-symmetric relation R is a not-symmetric relation, but a not-symmetric relation need not be anti-symmetric. If R is anti-symmetric on set X , then, by definition, for every pair $a, b \in X$, and $a \neq b$, if $(a, b) \in R$, then (b, a) fails to be in R . It implies for at least one pair $a, b \in X$, and $a \neq b$, if $(a, b) \in R$, then (b, a) fails to be in R , which implies R is a not-symmetric relation. For the converse, only an example is required: The relation $R = \{(a, b), (a, c), (c, a)\}$ is not-symmetric as $(a, b) \in R$, but (b, a) fails to be in R . However, R is not anti-symmetric as $(a, c) \in R$, and also $(c, a) \in R$.

Check Your Progress-1

1) Let R be a relation from $X = \{1, 2, 3, 4\}$, to $Y = \{1, 2, 3, 4\}$, with

$$R = \{(1, 2), (2, 1), (2, 3), (3, 1), (3, 2), (3, 4), (4, 3)\}$$

i) Give Graphic representation of R

ii) Give Matrix representation of R

.....

.....

.....

2) For each of the following relations, tell whether it is reflexive, not-reflexive or anti-reflexive

i) The relation of “is-congruent-to” on the set X of all triangles

ii) the relation “less-than-or-equal-to ($\leq$)” on numbers

iii) The relation “less-than ($<$)” on N

.....

.....

.....

3) For each of the following relations, tell whether it is reflexive, not-reflexive or anti-reflexive

i) $R = \{(1, 1), (2, 2), (1, 3), (2, 3)\}$ on $X = \{1, 2, 3\}$

ii) $R = \{(1, 2), (2, 1), (1, 3), (3, 1)\}$ on $X = \{1, 2, 3\}$

iii) $R = \{(1, 2), (2, 3), (1, 3)\}$ on $X = \{1, 2, 3\}$

3) Transitive, Not-Transitive, Anti-Transitive

a) A relation R on a set X is **transitive** if whenever $(a, b) \in R$ and also $(b, c) \in R$ —for elements $a, b, c \in X$; then $(a, c) \in R$.

b) From (a) above, a relation R on a set X is **not transitive** if—for some $a, b, c \in X$ —whenever $(a, b) \in R$ and also $(b, c) \in R$; then (a, c) fails to be in R .

Remarks 3:

If $(a, b) \in R$ and there is no $c \in X$ such that $(a, c) \in R$, then R is transitive

Some examples:

- i) Each of the relations $=$, and $<$ is transitive on set of numbers.
- ii) The relation is-congruent-with on the set of all triangles is transitive.
- iii) The relation is-mother-of is not transitive on the set of all human beings (because, if a is mother of b , and b is mother of c , then a is grand-mother of c , and a is not mother of c .)
- iv) The relation is-brother-of is transitive on the set of all human beings.

For (v) & (vi), on the set $X = \{1, 2, 3\}$,

- v) Each of the relations $R_1 = \{(1, 2), (3, 3), (2, 3), (1, 3)\}$, and $R_2 = \{(1, 1), (3, 3)\}$ is transitive,
- vi) The relation $R_3 = \{(1, 2), (3, 3), (2, 3), (2, 2), (2, 1)\}$ is not transitive, because, for $(1, 2) \in R$ and $(2, 3) \in R$, $(1, 3) \notin R$. The proof is completed. However, the claim could also be established in another way as follows: $(1, 2) \in R$, $(2, 1) \in R$, but $(1, 1) \notin R$.

c) A relation R on a set X is **anti-transitive** if for each triple $a, b, c \in X$; whenever $(a, b) \in R$ and also $(b, c) \in R$, then (a, c) fails to be in R .

Examples of anti-transitive:

- i) In the plane geometry, the relation in the set of all straight lines of being perpendicular ($\perp$) is anti-transitive: because, if for any three straight lines a, b , and c , with $a \perp b$, and $b \perp c$, then $a \not\perp c$ (a is not perpendicular to c)
- ii) By the same argument discussed earlier, the relation of is-mother-of on the set of human beings is anti-transitive.
- iii) On the set $X = \{1, 2, 3\}$, the relation $R = \{(1, 2), (2, 3), (3, 1)\}$ is anti-transitive, because only pairs of the form (a, b) and (b, c) are:
 - a) $(1, 2), (2, 3)$, for which $(1, 3)$ fails to be in R

b) (2,3), (3, 1), for which (2, 1) fails to be in R

c) (3, 1), (1, 2), for which (3, 2) fails to be in R.

Remarks 4:

a) Difference between not-transitive and anti-transitive: For a relation R to be not-transitive, we require for some triple $a, b, c \in X$ —if $(a, b) \in R$ and also, $(b, c) \in R$ then (a, c) fails to be in R. For a relation R to be anti-transitive, we require for every triple $a, b, c \in X$ —if $(a, b) \in R$ and also $(b, c) \in R$ then (a, c) fails to be in R.

b) Every anti-transitive relation is a not-transitive relation, but the converse may not be true. From (a) above, if R is anti-transitive, then for every triple $a, b, c \in X$ —if $(a, b) \in R$ and also $(b, c) \in R$ then (a, c) fails to be in R. Hence, for relation R, for at least one triple $a, b, c \in X$ —if $(a, b) \in R$ and also $(b, c) \in R$ then (a, c) fails to be in R. And hence R is not-transitive. For the opposite direction, just one example is sufficient. Example: (i) $R = \{(1, 2), (2, 3), (1, 3), (3, 2)\}$ on the set $\{1, 2, 3\}$ is a not-transitive relation as $(2, 3) \in R$, $(3, 2) \in R$, but $(2, 2) \notin R$. But it is not anti-transitive, as $(1, 2) \in R$, $(2, 3) \in R$, and also $(1, 3) \in R$.

4) A Simple Method of Generating Examples and Counter-Examples

Start with a small set say, $X = \{a, b, c\}$, and for the required relation R, take pairs, one by one, of elements of X, which satisfy, or violate the conditions as required for the relation R. For example, if R is required to be (i) reflexive & (ii) transitive, but not symmetric. Take pairs which make the relation reflexive, by including in R the pairs (a, a) , (b, b) and (c, c) . So far, the relation R is transitive and symmetric. But we require R to be nonsymmetric. Include, say, only (a, b) , but not (b, a) , so that $\{(a, a), (b, b), (c, c), (a, b)\} \subseteq R$.

As L. H. S. satisfies the required conditions, the process may be terminated.

Care must be taken that while satisfying one condition, some earlier satisfied condition may be violated. Then the process needs to be continued. For example, we wish to have a relation which is reflexive, but not symmetric and not transitive. Then we may continue from $\{(a, a), (b, b), (c, c), (a, b)\} \subseteq R$, and add a pair so that transitivity is disturbed, e. g. $\{(a, a), (b, b), (c, c), (a, b), (b, c)\} \subseteq R$, which satisfies the required conditions. If R is required not to be reflexive, exclude, say (a, a) from R. And so on.

Various combinations of the above-mentioned properties lead to some mathematically interesting and practically useful properties of relations. These interesting and useful combinations of properties include

- i) Equivalence of relations, and
- ii) Partial ordering of relations.

These two form the subject matter of next two sections.

2.4 EQUIVALENCE RELATIONS, AND PARTITION OF A SET

A relation R on a set X is said to be an equivalence relation, if R is reflexive, symmetric and transitive. In more detailed form, R is an equivalence relation on set X , if and only if

- (i) R is reflexive on X , i. e. for every $x \in X$, we have $(x, x) \in R$, and
- (ii) R is symmetric on X , i. e. for $x, y \in X$, if $(x, y) \in R$, then $(y, x) \in R$, and
- (iii) R is transitive on X , i. e. for $x, y, z \in X$, if $(x, y) \in R$, and $(y, z) \in R$, then $(x, z) \in R$

Some examples:

- i) The relation of equality ($=$) on the set of numbers, say Q , the set of rational numbers, is an equivalence relation
- ii) The relation of 'is-parallel-to' on the set of all straight lines in a plane, is an equivalence relation
- iii) The relation of 'is-congruent-to' on the set of all triangles in a plane, is an equivalence relation
- iv) The relation of 'is-sibling-of (male or female having same parents)' on the set of human beings, is an equivalence relation.

Not-equivalent relation: A relation R on a set X is **not an equivalence relation** if R

- i) fails to be reflexive on X : i. e., even for one particular $a \in X$, if $(a, a) \notin R$. For example, for set $X = \{1, 2, 3\}$, the relation $R = \{(2, 2), (3, 3), (2, 3), (3, 2)\}$ is not equivalence relation, as it is not reflexive, because $(1, 1) \notin R$, despite $(2, 2), (3, 3) \in R$ (this is an example of a relation which is symmetric, and transitive, but which is not reflexive)

Or

- ii) fails to be symmetric on X , i. e. even for one pair of elements $a, b \in X$, we have $(a, b) \in R$, but $(b, a) \notin R$. For example, $X = \{1, 2, 3\}$, the relation $R = \{(1, 1), (2, 2), (3, 3), (2, 3)\}$ is not symmetric, and hence not equivalence relation on X , as $(2, 3) \in R$, but $(3, 2) \notin R$. (this is an example of a relation which is reflexive, and transitive, but which is not symmetric)

Or

- iii) fails to be transitive on X , i. e. for $a, b, c \in X$, if for pair of elements $(a, b) \in R$, and $(b, c) \in R$ but $(a, c) \notin R$. For example, $X = \{1, 2, 3\}$, the relation $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1), (2, 3), (3, 2)\}$ is not transitive, and hence not equivalence relation on X , as $(1, 2)$, and $(2, 3) \in R$, but $(1, 3) \notin R$. (this is an example of a relation which is reflexive, and symmetric, but which is not transitive)

Next, we discuss a concept which is quite closely associated with equivalence relation, viz Partition of a set. The close relation (which we will prove) is that for each equivalence relation R on set X , there is a partition of X , and for every partition of a set, say X there is an equivalence relation on X .

2.4.1 Partition of A Set

For a given set X , a partition of X is a non-empty, finite or infinite, collection $P = \{X_i\}$, where X_i 's are mutually disjoint non-empty subsets of X , with their union equal to whole X . More explicitly, a collection $P = \{X_i\}$ is a partition of X , if

- i) $X_i \neq \emptyset$, for each $X_i \in P$,
- ii) $X_i \cap X_j = \emptyset$, for every distinct pair of X_i and X_j in P
- iii) $\bigcup_{i \in P} X_i = X$

A non-empty set may have several distinct partitions. Examples: Let $X = \{a, b, c\}$. Then each of the following is a partition of X :

- i) $\{\{a\}, \{b\}, \{c\}\}$; (ii) $\{\{a\}, \{b, c\}\}$ (iii) $\{\{a, b\}, \{c\}\}$ (iv) $\{\{b\}, \{a, c\}\}$ (v) $\{\{a, b, c\}\}$

2.5 EQUIVALENCE CLASS OF AN EQUIVALENCE RELATION

For an equivalence relation R on a set X , the subset $[a]$ of X defined as $[a] = \{x \in X : (x, a) \in R\}$ is called an equivalence class of a under the equivalence relation R .

Example: Consider the relation $R = \{(a, a), (b, b), (b, c), (c, b), (c, c)\}$ on set $X = \{a, b, c\}$. The relation is an equivalence relation. Two equivalence classes in X under R are $\{a\}$ and $\{b, c\}$.

Every equivalence relation on a set X determines a partition of X . Conversely, if we are given a partition of a non-empty set, then it determines an equivalence relation on X .

Let $P = \{A_i\}$ be a partition of X . Then we define a relation R on X such that for $x, y \in X$, $(x, y) \in R$ if and only if x and y belong to the same class A_i of the partition. Then, to show R is an equivalence relation on X : (I) Reflexive: for each $x \in X$, the elements x , and x belong to the same class, hence $(x, x) \in R$. (II) Symmetry: if $(x, y) \in R$, i. e., if x and y belong to the same class of partition, then y and x also belong to the same class. Hence, $(y, x) \in R$. (III) Transitivity: if $(x, y) \in R$, and $(y, z) \in R$. Then, by definition of R , x and y are in the same class A_i and y and z are in the same class A_j . By definition of partition, A_i and A_j are either equal or are disjoint. But y belongs to both A_i and A_j . Hence, $A_i = A_j$ implying x and z belong to the same class of the same class of the partition; implying $(x, z) \in R$.

2.5.1 Applications of Equivalence Relation and Partition of a Set

Motivation for equivalence relation is that for some specific purpose, many apparently distinct elements behave identically, and hence need not be distinguished from each other. By clubbing all such elements, we reduce complexity for designing solutions of problems concerning the purpose. For example, for designing a timetable for a school with 1000 students, 5 classes (say, VIII to XII), each class having 4 sections (say, A, B, C & D), and each section having 50 students. Now the timetable need not be designed independently for 1000 distinct students, but only for 20 class sections (5×4). For the purpose of designing a timetable, all the 50 students of a class-section are treated as a single entity, which is an equivalence class under the equivalence relation R with $(x, y) \in R$ if and only if students x and y belong to the same class-section. These 20 class sections determine a partition of the set of all students in the school.

The equivalence relation has useful applications in the design of digital computers and other sequential machines.

2.5.2 Partial Order Relation, Partially Ordered Sets

A relation R on a set X is said to be a partial order, if R is reflexive, antisymmetric and transitive. In more detailed form, R is an partial order on set X , if and only if

- i) R is reflexive on X , i. e. for every $x \in X$, we have $(x, x) \in R$, and
- ii) R is anti-symmetric on X , i. e. for $x, y \in X$, if $(x, y) \in R$, and $(y, x) \in R$, then $x=y$, and
- iii) R is transitive on X , i. e. for $x, y, z \in X$, if $(x, y) \in R$, and $(y, z) \in R$, then $(x, z) \in R$

Some examples:

- i) The relation of less-than-or-equal-to ($\leq$) on the set of numbers, say Q , the set of rational numbers, is a partial order relation
- ii) The relation of subset-of ($\subseteq$) on a set of subsets of a given set, is a partial order.
- iii) The relation of is-divisor-of on N , the set of natural numbers is a partial order.

The relation is called partial order because, some pairs of elements of domain under a partial order may not be related. For example, in (ii) just above, the two subsets $\{a, b, c\}$ and $\{b, c, d\}$ are not related to each other under subset-of ($\subseteq$) relation. Similarly, for (iii) just above, the number 5 does not divide 9, nor 8 divides 5. Hence, 5 and 9 are not related to each other under divisibility relation.

Partially Ordered Set (Poset): An ordered pair (X, R) - where X is non-empty set and R , a partial order relation on X - is called a partially ordered set (for short, Poset). For example, $(N, \text{is-divisor-of})$ is a partially ordered set.

Total order: For a partial order R on a set X , if for every pair $x, y \in X$, either $(x, y) \in R$, or $(y, x) \in R$, i. e. every pair of elements is related, then the relation R is called Total order, and (X, R) is called a totally ordered set. For example, If for the relation

Relations

is-divisor-of is defined on the set $\{2^0, 2^1, 2^2, 2^3, \dots\}$, of all powers of 2, then is-divisor-of is a total order on the set. In general, the relation is-divisor-of defined on the set $\{p^0=1, p^1, p^2, p^3, \dots\}$, of all powers of p , for a prime number p , is a total order.

Hasse Diagram

In partial order theory, Hasse Diagram is a graphical representation of a finite partially ordered set obtained as follows. Let (X, R) be a finite partially ordered set. The Hasse diagram is obtained by drawing an upward line from x to y whenever for $x, y \in X$, we have $x R y$, (i.e., $(x, y) \in R$) as shown and explained in the following examples.

Example: Let $X = \text{divisors of } 12 = \{1, 2, 3, 4, 6, 12\}$, and for $x R y$ if x is a proper divisor of y , for $x, y \in R$. Then the corresponding Hasse diagram is shown in Fig. 2.5.

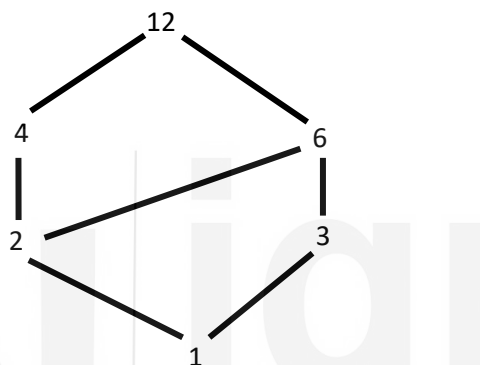


Fig. 2.5: Hasse Diagram - 1

Example: Let $X = \text{Power set of } \{x, y, z\}$, i. e. $X = \{\emptyset, \{x\}, \{y\}, \{z\}, \{x, y\}, \{x, z\}, \{y, z\}, \{x, y, z\}\}$. Define R on X such that for $A, B \in X$, then $(A, B) \in R$, if $A \subset B$. The Hasse diagram is as shown in Fig 2.6.

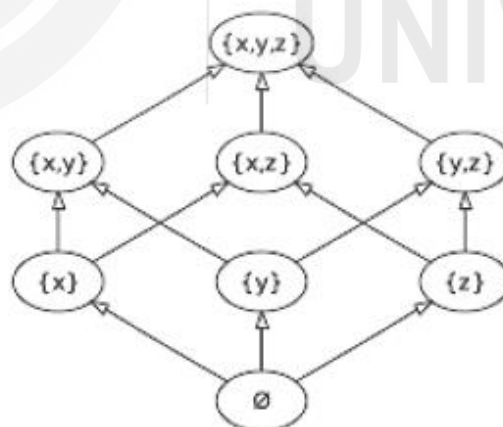


Fig. 2.6: Hasse Diagram - 2

Example: Let $X = \{3, 4, 12, 24, 48, 72\}$, and under the relation R , for $x, y \in X$, $(x, y) \in R$, if x is proper divisor of y . The Hasse diagram is shown in Fig 2.7.

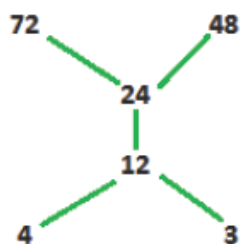


Fig. 2.7: Hasse Diagram -3

Following is an another example of a totally ordered set $(\{1,2,3,4,8\},/)$, where “/” is division symbol.

As we know a set is "totally ordered" if **every pair** of elements in the set is comparable. This means for any two numbers a and b , above either a divides b or b divides a .

Hasse diagram for this set will be a strait line.



Figure 2.8 Hasse Diagram - 4

Some Definitions

Here we provide definitions of maximal element and minimal element, of a set under a partial order.

In the previous Fig. 2.7 there are two elements, viz. 4 and 3, at the lowest level in the sense that $x \in X$ does not divide 3, for $x \neq 3$, and $x \in X$ does not divide 4 for $x \neq 4$. Such elements are called minimal elements.

Thus, in a poset (X, R) , an element a of X is a minimal element under R , if there does not exist x in X , such that x is related to element a . A Poset may have one, two or more minimal elements. We have shown in a previous example that there are cases with two minimal elements. When we have only one minimal element, we also call it the minimum element; in this case, it is called the **least element**.

Maximal Element: In Figure 2.7, there are two elements, viz. 48 and 72, at the highest level in the sense that 48 does not divide $x \in X$, for $x \neq 48$, and also 72 does not divide $x \in X$ for $x \neq 72$. Such elements are called Maximal elements.

Thus, in a Poset (X, R) , an element a of X is a maximal element under R , if there does not exist y in X , such that a is related to y . A Poset may have, one, two or more maximal elements. In the example corresponding to Fig.3.7, we have two maximal elements, viz. 48 and 72.

In the example corresponding to Fig.2.5 we have just one maximal element, viz. 12. In such cases, the maximal element is also called greatest element of the Poset.

Note: Point to remember that **least and greatest element are unique.**

Check Your Progress -2

1) For each of the following relations, tell whether it is transitive, not-transitive, anti-transitive.

i) In the plane geometry, the relation R —in the set of all straight lines—is defined as $R(a, b) \in R$, for straight lines a and b , if and only angle between a and b is 30 degrees.

ii) The relation less-than-or-equal-to ($\leq$) on numbers

iii) The relation is-brother-of, if underlying set X = set of all human beings

.....

.....

.....

.....

2) Show that the three properties of relations, viz. reflexive, symmetric and transitive are independent of each other, i. e. a relation may be (i) reflexive, but neither symmetric, nor transitive; (ii) symmetric, but neither reflexive, nor transitive; and (iii) transitive, but neither reflexive, nor symmetric.

.....

.....

.....

.....

3) Find out partition of the set $X = \{1, 2, 3, 4\}$ given by the equivalence relation

$$R = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 3), (3, 1), (2, 4), (4, 2)\}$$

.....

.....

.....

.....

2.6 OPERATIONS ON RELATIONS

A relation on input of required number of arguments returns true/false, e. g. is- less-than (3, 5) is true, and is-less-than (5,3) is false. On the other hand, an operation on input of required number of arguments returns a new element of the domain, e. g. Sum (3, 5) returns 8.

In this section, we discuss operations on relations, i. e., operations for which inputs are relations, and the output is again a relation. Unless otherwise mentioned, all operations are binary relations on (a single) set, say, X .

The operations which we discuss may be classified as:

- a) Set-type operations: union, intersection and difference of relations
- b) Mapping-type operations: inverse of operation, composition
- c) Closure operations: reflexive-closure, symmetric-closure, transitive closure.

2.6.1 Set-type Operations on Relations

The set-type operations on relations include union of relations, intersection of relations etc. The definitions of these operations are exactly on the lines of the definitions of operations on sets given in Unit 1 of this course. We explain this type of operation on relations only through examples. For the discussion of this type of operations on relations, let R and S be two relations from $X = \{1, 2, 3, 4\}$ to $Y = \{a, b, c\}$, with

$$R = \{(1, a), (2, a), (2, c), (3, b), (4, b), (4, c)\}$$

$$S = \{(1, b), (2, a), (3, a), (3, b), (4, a)\}$$

Then, Universal relation U has $4 \times 3 = 12$ elements of the form (x, y) with all $x \in X$, and all $y \in Y$.

i) Union of R and S denoted by $R \cup S$, is given by:

$$R \cup S = \{(1, a), (2, a), (2, c), (3, b), (4, b), (4, c), (1, b), (3, a), (4, a)\}$$

ii) Intersection of R and S denoted by $R \cap S$, is given by $R \cap S = \{(2, a), (3, b)\}$

iii) Difference $R - S = \{(1, a), (2, c), (4, b), (4, c)\}$, and

$$\text{Difference } S - R = \{(1, b), (3, a), (4, a)\}$$

iv) Complement of relation $R = R' = \{(1, b), (1, c), (2, b), (3, a), (3, c), (4, a)\}$.

2.6.2 Mapping-type operations: inverse of operation, composition

i) For a relation R from set X to Y , the inverse relation of R , denoted by R^{-1} is the relation from Y to X , such that $(y, x) \in R^{-1}$, if and only if $(x, y) \in R$. For example, if $R = \{(1, a), (2, a), (2, c), (3, b), (4, b), (4, c)\}$ from $X = \{1, 2, 3, 4\}$ to $Y = \{a, b, c\}$ then $R^{-1} = \{(a, 1), (a, 2), (c, 2), (b, 3), (b, 4), (c, 4)\}$ is from $Y = \{a, b, c\}$ to $X = \{1, 2, 3, 4\}$. Note the difference between R' and R^{-1} .

ii) Composition of relations denoted by $R \circ S$.

Here, R and S are defined in a slightly different way than these were defined just above.

Relations

Let R be a relation from $X = \{1, 2, 3, 4\}$ to $Y = \{a, b, c\}$, say, $R = \{(1, a), (2, a), (2, c), (4, c)\}$;

And, S be a relation from $Y = \{a, b, c\}$ to $Z = \{\text{triangle}, \text{square}\}$, say $S = \{(a, \text{triangle}), (b, \text{square}), (b, \text{triangle}), (c, \text{square})\}$. Note the Codomain of $R = \text{Domain of } S$, is essential for the definition of composition.

Then the relation $R \circ S$ is from $X = \{1, 2, 3, 4\}$ to $Z = \{\text{triangle}, \text{square}\}$, given by

$R \circ S = \{(1, \text{triangle}), (2, \text{triangle}), (2, \text{square}), (4, \text{square})\}$

2.6.3 Closure Operations

The reflexive-closure, symmetric-closure, and transitive closure operations are relevant only for a relation R on a set, say X . The operations are illustrated through examples:

Let $X = \{1, 2, 3\}$, and $R = \{(2, 2), (3, 3), (2, 3), (3, 1)\}$

i) Reflexive-closure of a relation R is a relation S such that $R \subseteq S$, and S is obtained from R by adding minimum number of ordered pairs (x, x) so that S is reflexive. For example, for $R = \{(2, 2), (3, 3), (2, 3), (3, 1)\}$, Reflexive-closure $(R) = \{(2, 2), (3, 3), (2, 3), (3, 1), (1, 1)\}$

ii) Symmetric-closure of a relation R is a relation S such that $R \subseteq S$, and S is obtained from R by adding minimum number of ordered pairs (x, y) so that S is symmetric. For example, for $R = \{(2, 2), (3, 3), (2, 3), (3, 1)\}$, Symmetric-closure $(R) = \{(2, 2), (3, 3), (2, 3), (3, 1), (3, 2), (1, 3)\}$

iii) Transitive-closure of a relation R is a relation S such that $R \subseteq S$, and S is obtained from R by adding minimum number of ordered pairs (x, y) so that S is transitive. For example, for $R = \{(2, 2), (3, 3), (2, 3), (3, 1)\}$, Transitive-closure $(R) = \{(2, 2), (3, 3), (2, 3), (3, 1), (2, 1)\}$.

Check Your Progress - 3

1) Under the relation of is-proper-divisor-of, draw Hasse diagram for set X , where $X =$ all divisors of

(i) 30 (ii) 24 and for each case, write all the minimal elements, and maximal elements.

.....
.....
.....

2) Under the relation of *is-proper-divisor-of*, draw Hasse diagram for set $X = \{1, 2, 3, \dots, 10\}$, and tell all the minimal elements, and maximal elements.

.....
.....
.....

2.7 SUMMERY

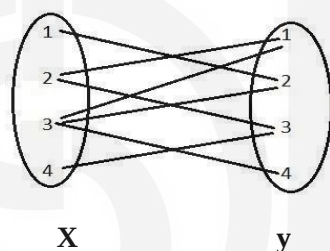
Relation one of the most fundamental concepts of Mathematics. In this unit, we introduced the concepts through respective definitions and illustrative examples. Also, we discussed three main properties of relations, viz. reflexivity, symmetry, and transitivity, along with some variants of these properties and some interesting combinations of these properties, viz. equivalence relations and partial orders. Finally, we discussed some operations on relations.

2.8 ANSWER TO CHECK YOUR PROGRESS

Check Your Progress – 1

1.

i) Graphic Representation



ii) Matrix Representation

	1	2	3	4
1	0	1	0	0
2	1	0	1	0
3	1	1	0	1
4	0	0	1	0

2. i) The relation of is-congruent -to on set X of all triangles is **symmetric**.

ii) The relation less-than-equal to ($\leq$) on numbers is not symmetric, as $2 \leq 3$ is true but $3 \leq 2$ is not true. Also, $\leq$ is anti-symmetric as $x \leq y$ and $y \leq x$ then $x = y$.

iii) The relation $<$ on $\mathbb{N}$ is asymmetric.

3. i) $\{(1,1), (2,2), (3,3)\}$ is both symmetric and antisymmetric.

ii) $\{(1, 2), (2,3), (1,3)\}$ is antisymmetric, but not symmetric.

iii) $\{(1,2),(2,1),(1,3),(3,1)\}$ is symmetric, but not anti-symmetric, as $(1,2), (2,1)$, but $1 \neq 2$.

Check Your Progress – 2

1.

i) The relation R defined as $R(a, b)$, for straight lines a and b , if and only angle between a and b is 30 degrees, is not *transitive*, because for $R(a, b)$ and $R(b, c)$, the angle between (lines a and b) and also (lines b and c) may be 30 degrees, but angle between a and c may be 60 degree also.

ii) The relation less-than-or-equal-to ($\leq$) on numbers is *transitive*

iii) The relation *is-brother-of*, if underlying set X = set of all human beings is *transitive*.

2. The three properties of relations on a set X , are independent of each other. Examples of relation which is:

i) *reflexive, but neither symmetric, nor transitive*:

R on Z , the set of integers, $(a, b) \in R$, if and only if $0 \leq a-b \leq 1$

Reflexive: $0 = a-a \leq 1$; **Non-symmetric:** $(4,3) \in R$, $(3,4) \notin R$;

nor transitive $(4,3) \in R$, $(3,2) \in R$, $(4,2) \notin R$

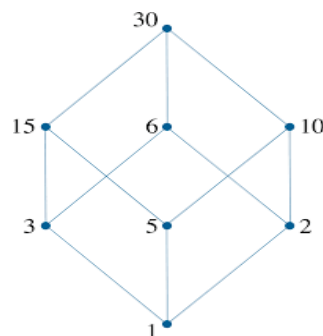
ii) symmetric but neither reflexive, nor transitive: R , the relation of being perpendicular to each other in the set of all straight lines in a plane.

iii) R is less than ($<$) on the set of numbers, say, rational numbers

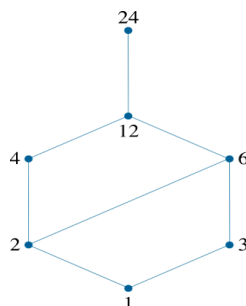
3. Partition of $X = \{ \{1,3\}, \{2,4\} \}$

Check Your Progress – 3

1. i)



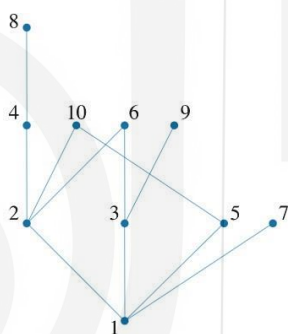
For(i) above, Minimal element = 1, Maximal element = 30;



ii).

For(ii) above, Minimal element = 1, Maximal element = 24

2. Minimal element=1, Maximal elements=5,6,7,8,9,10.



2.9 REFERENCES AND FURTHER READINGS

1. Discrete Mathematical Structures By Bernard Kolman , Robert C. Busby , Sharon Cutler Ross , PHI 2002
2. Mathematics Class XI By RD Sharma , Dhanpat Rai Publication , 2011
3. Mathematics Class XI By Ravi Prakash , GRB Books, Edition 2025
4. MCS-013: Discrete Mathematics
Unit-1 Sets, Relations and Functions
(<https://egyankosh.ac.in/handle/123456789/1140>)
5. MEC-203 : Quantitative Methods
Unit-3 Relations and Functions
(<https://egyankosh.ac.in/handle/123456789/98141>)
6. MST-001: Foundation in Mathematics and Statistics
Unit-2 Functions
(<https://egyankosh.ac.in/handle/123456789/92122>)